Component Selection:

Component	Cost	Link	Units	Size
Buck IC	$1.61 for 1 cut tape	https://www.digikey.com/en/products/detail/texas-instruments/TPS565201DDCR/7732455 *	1	**SOT-23-6 Thin, TSOT-23-6**
Inductor	$1.31 for 1 cut tape	https://www.digikey.com/en/products/detail/bourns-inc/SRP7050AA-3R3M/13910669?utm_source=chatgpt.com	1	0.287" L x 0.260" W (7.30mm x 6.60mm)
Capacitor	0.1 microfarad $0.14 for cut tape output capacitor: $0.75 for cut tape input capacitor: $0.30	https://www.digikey.com/en/products/detail/tdk/C1608X8R1E104K080AA/927126(bootstrap capacity? Really low capacitance, good for not big currrent) https://www.digikey.com/en/products/detail/tdk/C3216X5R1C226M160AB/2443479?gclsrc=aw.ds&gad_source=1&gad_campaignid=24216361874&gbraid=0AAAADrbLliDh-ePJIdfGPBPtffdaltxe&gclid=CjwKCAjw88HWBhA3EiwAo2Zd6K_Ppj_Rs6WADfvjctOrJZSiqE3L2oG64XnmOmcwtghhN3tE4zmtdxoCuQcQAvD_BwE (output capacitor) https://www.digikey.com/en/products/detail/murata-electro	3 note: the packaging for this capacitor is bigger than the recommended on the onboarding sheet however the datasheet from the manufacturer recommended to use this one so i choose it.	0.063" L x 0.031" W (1.60mm x 0.80mm) 0.126" L x 0.063" W (3.20mm x 1.60mm) 0.063" L x 0.031" W (1.60mm x 0.80mm)

		nics/GRT188R61C106KE13D/5416739 (input capacitor)		
Total Cost	$4.11			

Options (ic):

https://www.digikey.com/en/products/detail/texas-instruments/TPS563208DDCR/5825114 (580mhz so not in target range)

https://www.digikey.com/en/products/detail/texas-instruments/TPS565201DDCR/7732455 (good but 5A current rating, more than the 2A but not out of range?)

https://www.digikey.com/en/products/detail/texas-instruments/TPS54202HDDCR/6026259 (input can take a lot so im actually not sure if thats better, kinda out of the target range tho)

https://www.digikey.com/en/products/detail/eaton-electronics-division/SDCL1V4020-4R7M-R/14564407 (inductor current rating is lower)

- current rating tells you how much electrical current that a component can carry safely without burning or overheating, so higher current rating should be better because in the case that more current goes through the circuit/wire it can handle more (?)
- for the integrated circuit that i chose, the pins are:
    - GND (pin 1): connect to ground.
    - switch node (pin 2): connect to one side of the inductor.
    - V in (pin 3): connect to the input supply.
    - voltage feedback (pin 4): connect to the midpoint of the feedback resistor divider, which senses the output voltage.
    - enable (pin 5): the on and off control pin. pull it high to turn the converter on.
    - VBST (pin 6): bootstrap pin? It works with a smaller capacitor 0.1 µF thats connected between this pin and the switch node to help the chip switch its internal transistor

resistor selection:

https://www.digikey.com/en/products/detail/stackpole-electronics-inc/RNCF0805TKY10K0/2269701 (r2)

https://www.digikey.com/en/products/detail/koa-speer-electronics-inc/RK73H1JTTD1002F/9844954 (actual r2, other one packaging was too big and i did not realize)

https://www.digikey.com/en/products/detail/vpg-foil-resistors/Y161155K7800A9R/4229840?s=N4IgjCBcoEwAwA4CsVQGMoDMCGAbAzgKYA0IA9lANohJIB0A7AgAQDWA8gBYC2%2BIAuqQAOAFyggAqgDsAliPaYAsoWz4ArgCdCIAL47SMKiC34Z%2BEWQ3MBOoA (r1) (not the chosen one)

- resistors with higher tolerance deviate more from the value so choosing a resistor thats rounder up but has lower tolerance is better.

https://www.digikey.com/en/products/detail/koa-speer-electronics-inc/RK73H1JTTD5602F/9845000 (r1) (one that is used in schematic)